

Africa Countries — Analytical Report

POPULATION · TERRITORY · LINGUISTIC COMPLEXITY

A data-driven analysis of 54 African countries combining demographic weight, territorial extent, and linguistic diversity — powered by REST API, Python, SQL, and Power BI.

54 Countries

Across 5 African regions

14 Variables

After feature engineering

10 SQL Queries

Decision-making dimensions

1 Dashboard

Power BI interactive visuals



Dataset Summary

DATA SOURCE: AFRICA COUNTRIES REST API

Raw Dataset

54 African countries

9 raw variables → 14 final variables after feature engineering

Static data — no temporal dimension

Format: JSON → CSV

Variable Categories

Geographic

country, african_region, area_km2, is_landlocked

Demographic

total_population, population_density, population_rank

Linguistic

language_count, currency_count

Engineered

area_category, linguistic_diversity_level, complexity_score

- ❑ **Data Quality:** No missing values detected. Outliers retained (Nigeria, Algeria, DRC) — valid geographic realities. Boolean converted to TINYINT(1) for SQL/Power BI compatibility.

Python EDA Pipeline

A four-step reproducible pipeline from raw API data to a clean, analysis-ready dataset exported to MySQL.



Step 1 — API Collection

HTTP GET request from Africa Countries API. JSON parsed into pandas DataFrame, exported to CSV.



Step 2 — Data Quality Audit

Variable types verified. IQR outlier detection. Landlocked bool → int. Columns renamed to snake_case.



Step 3 — Feature Engineering

6 new variables: `population_density`, `population_rank`, `area_category`, `linguistic_diversity_level`, `connectivity_profile`, `complexity_score`.



Step 4 — MySQL Export

54 rows × 14 columns imported via SQLAlchemy + pymysql using a single `to_sql()` operation.

Key Engineered KPI

The **`complexity_score`** is the central KPI — a normalized index (0–100) combining population, territory, and linguistic diversity into a single strategic measure for each country.

Connectivity Profile

A geopolitical openness indicator combining landlocked status and border count — enabling nuanced trade and logistics analysis beyond simple geographic classification.

SQL Analysis: Demographic Power

Q1 & Q2 — POPULATION QUERIES

Q1 — Top 10 Most Populated Countries

1 Nigeria — #1

223.8 million
inhabitants · Density:
242 inh/km²

2 DR Congo — #2

112.8 million · Ethiopia
#3 at 111.6M

3 Uganda — Highest Density

190 inh/km² in the top
10 — extreme
demographic pressure

- These 3 countries
alone represent a
disproportionate
share of continental
population.

Q2 — Region with Highest Total Population

Western Africa

440M inhabitants across
17 countries

Eastern Africa

440M across 19
countries · Highest avg
density: 233 inh/km²

Southern Africa

Least populated: only
88.9M inhabitants





SQL Analysis: Territorial Dominance

Q3 & Q4 — AREA QUERIES

Q3 — Top 10 Largest Countries by Area

Algeria — #1

2.38 million km² ·

Category: Very Large ·
Dominates Northern
Africa

DR Congo — #2

2.34M km² · Sudan follows
at 1.87M km²

All Top 10

Fall in the "Very Large"
category — presenting
major logistics and
infrastructure challenges

Q4 — Region with Largest Total Area

Northern Africa

7.9M km² for only 7
countries · Avg: 1.13M
km²/country

Middle Africa

7.2M km² across 10
countries

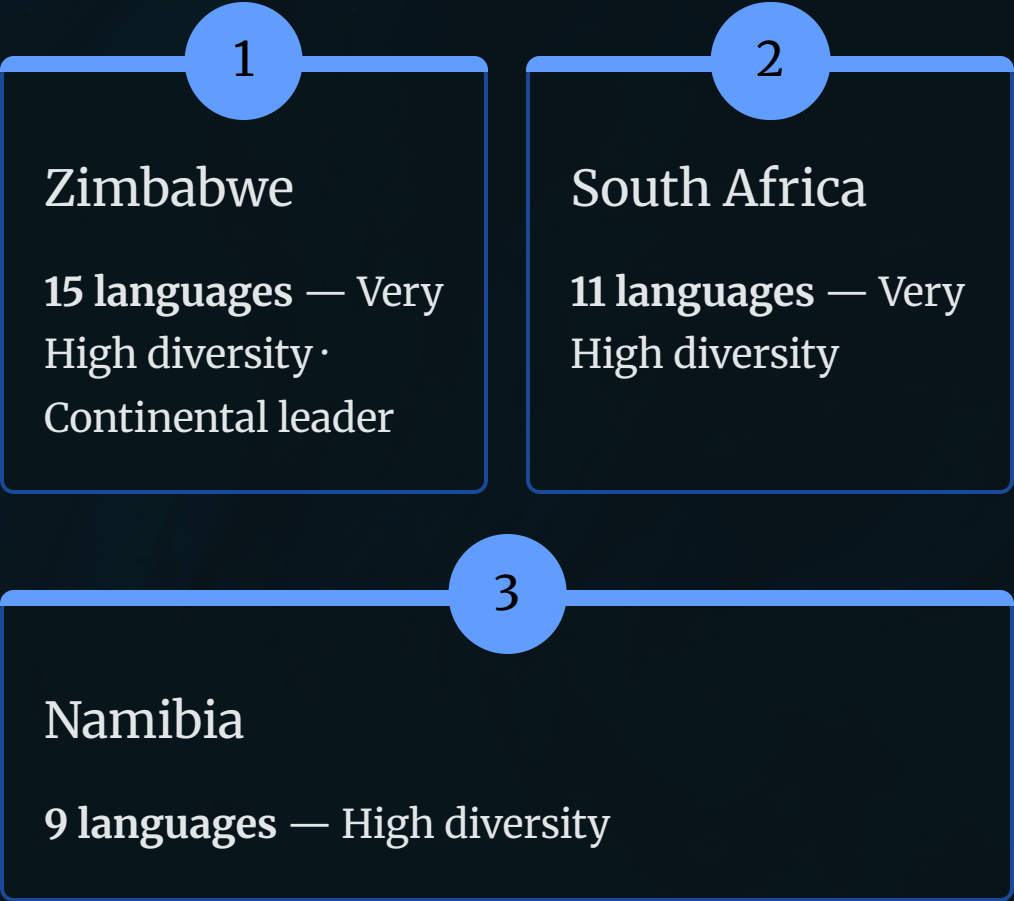
Eastern Africa

Most countries (19) but
**smaller average
territory**

SQL Analysis: Linguistic Diversity

Q5 & Q6 — LANGUAGE QUERIES

Q5 — Countries with Highest Language Count



❑ Most African countries have only 1–3 languages (Low level). Very high linguistic diversity is concentrated in very few countries.

Q6 — Region with Highest Average Linguistic Diversity



Southern Africa

Highest average — counter-intuitive result, higher than Central or West Africa



Western Africa

Lowest average languages per country despite large number of nations

Southern Africa max: 15 languages (Zimbabwe)

SQL Analysis: Complexity Score

Q7 & Q8 — STRATEGIC COMPLEXITY QUERIES

The **complexity_score** (normalized 0–100) is the central KPI combining population, territory, and linguistic diversity into a single strategic index.

DR Congo

Score: 59.15

Population: 112.8M · Area: 2.34M km²

South Africa

Score: 50.3

11 languages · 63.1M inhabitants

Nigeria

Score: 46.26

223.8M inhabitants — highest population on continent

Zimbabwe

Score: 41.34

15 languages — highest language count on continent

- ❑ **Q8 — High Complexity + High Linguistic Diversity:** Only 3 countries cumulate both — South Africa (50.3 / Very High), Zimbabwe (41.34 / Very High), Namibia (31.05 / High). All 3 are in **Southern Africa** — confirming it as the most strategically complex region.

SQL Analysis: Geopolitical Connectivity

Q9 & Q10 — CONNECTIVITY & REGIONAL PROFILE QUERIES

Q9 — Landlocked Countries with Most Borders



Zambia — 8 Land Borders

Best-connected landlocked country · Profile:
Landlocked Connected



Niger & Mali — 7 Borders Each

Strong regional trade connectivity potential despite
no sea access

❑ Being landlocked does not mean geopolitical isolation. These countries have strong regional trade connectivity potential.

Q10 — Average Profile by African Region

Region	Key Metric	Value
Northern Africa	Avg Territory	1.13M km ²
Southern Africa	Avg Complexity	23.25
Eastern Africa	Avg Density	233 inh/km ²
Middle Africa	Avg Area	723K km ²
Western Africa	Countries	17 nations

Power BI Dashboard Overview

POPULATION, TERRITORY AND LINGUISTIC COMPLEXITY DASHBOARD

A single-page interactive dashboard structured across 3 reading levels — from high-level KPIs to granular country-level exploration.

54

Total Countries

African nations analyzed

1B

Total Population

Continental aggregate

13.67

Avg Complexity

Mean complexity score

27%

Landlocked Rate

Share of landlocked nations

2.17

Avg Languages

Per country average

Level 2 — Main Analytical Visuals

- Population & Area by Region — clustered bar chart
- Population vs Complexity Score — scatter plot (bubble size = area)
- Linguistic Diversity by Region — 100% stacked bar chart

Level 3 — Secondary Visuals & Slicers

- Top 5 Countries by Complexity Score — ranked bar chart
- Area Category Distribution — donut chart
- Top 5 Countries by Language Count — bar chart
- Connectivity Profile Distribution — bar chart

☐ Active Slicers: Connectivity profile · African region · Linguistic diversity level · Area category · Is_landlocked

Business Recommendations

5 STRATEGIC RECOMMENDATIONS FOR DECISION-MAKERS

1

Prioritize DR Congo, South Africa & Nigeria

These 3 countries dominate all 3 dimensions simultaneously. Any continental strategy must allocate dedicated resources to these markets first.

2

Integrate Multilingual Strategy for Southern Africa

Southern Africa has the highest average linguistic diversity (6.83 languages/country). All operations require structured multilingual communication and governance approaches.

3

Don't Exclude Landlocked Countries from Trade

Zambia, Niger and Mali have up to 8 land borders each. Their Landlocked Connected profile gives them strong regional trade potential despite having no sea access.

4

Leverage Northern Africa's Territorial Potential

Northern Africa dominates in land area (avg 1.13M km²) but has the lowest population density. High potential for energy, agriculture, and infrastructure investment.

5

Apply Country-Level, Not Regional Strategies

The complexity_score reveals deep heterogeneity within each region. Uniform regional strategies are ineffective — each country requires a differentiated approach based on its specific complexity profile.